



The Corruption Paradox: An Empirical Analysis of Credit Risk in South Asia

An Econometric and Machine Learning Approach (2014-2024)

Sumit Kothari B-062
Harsh Shukla B-040
Rudra Koli B-044
Kshitij Gaikwad B-007
Sidhesh Mohanty B-017

What are Non-Performing Loans (NPLs)?

- Loans where principal or interest payments are 90+ days past due.
- A primary indicator of a banking system's health and credit risk.

Why They Matter

- **Erode Profitability:** Reduce interest income and require higher provisions for losses, depleting bank capital.
- **Constrain Economy:** Limit a bank's ability to extend new credit, leading to a "credit crunch" that stifles investment.
- **Systemic Risk:** A surge in NPLs can signal an impending banking crisis.

Context: South Asia

- A region of dynamic growth but also macroeconomic volatility and institutional weakness.
- Many banking systems have seen a worrying rise in NPLs, threatening economic progress.

Research Question: What are the key bank-specific, macroeconomic, and institutional factors that determine the level of NPLs in South Asia?

1. Greasing the Wheels Theory

- Source: *Mauro (1995); Chen et al. (2013)*
- Concept: Suggests that corruption (through bribery) can accelerate administrative and financial processes, making it easier for firms to obtain loans. However, this can lead to moral hazard.

1.2. Sand in the Wheels Theory

- Source: *Cooray & Schneider (2018)*
- Concept: Proposes that corruption hinders the efficient functioning of the financial and banking systems. It leads to unfair resource allocation, inefficiency, and increased credit risk.

Study	Region	Key Findings
Bougatef (2016)	Emerging Markets	Corruption ↑ NPLs
Rehman et al. (2020)	Pakistan	Anti-corruption ↓ NPLs
Louzis et al. (2012)	Greece	GDP, inflation, and bank size affect NPLs
Thanh et al. (2025)	SE Asia	XGBoost shows corruption strongly predicts NPLs
Goel & Hasan (2011)	Cross-country	Weak governance ↑ credit risk

Nguyen Thanh, C., Huy, T. P., Pham Hong, T., & Bui Nguyen Quoc, A. (2025). Credit risk prediction with corruption perception index: machine learning approaches. *Cogent Business & Management*, 12(1), 2461731.

- Key Findings:

- This paper provides a complete blueprint for predicting credit risk in Southeast Asia using machine learning and corruption data.
- The XGBoost model performed best, achieving high accuracy (R^2 of 0.932).
- Bank Size and the Corruption Perceptions Index (CPI) were identified as the most significant predictors of credit risk.

- Relevance to Our Study:

- Offers a proven methodological roadmap (models, features, metrics) that we can adapt for the South Asian context.
- Establishes a clear performance benchmark for our models.

Rehman, A., Abdul Adzis, A., & Mohamed Arshad, S. B. (2020). The relationship between corruption and credit risk in commercial banks of Pakistan. *International Journal of Innovation, Creativity and Change*, 11(1), 701-715.

- Key Findings:

- In Pakistan, better "Control of Corruption" (COC) has a significant negative relationship with Non-Performing Loans (NPLs).
- Larger bank size was associated with higher credit risk, while higher profitability (ROA) and better capitalization reduced risk.

- Relevance to Our Study:

- Validates the research hypothesis within a key South Asian country.
- Provides a proven set of bank-specific and macroeconomic variables (e.g., bank size, ROA, GDP, inflation) for feature selection.

Jenkins, H. (2025). The Effect of Corruption on Banks'™ Credit Risk: A Comparative Analysis Between Islamic and Conventional Banks (No. 2024-08). *JDI Executive Programs*.

- Key Findings:

- Corruption has a significantly stronger positive link to credit risk in conventional banks compared to Islamic banks.
- Islamic banks show greater resilience, possibly due to profit-and-loss sharing models and internal Sharia committees acting as controls.

- Relevance to Our Study:

- Provides strong empirical evidence validating the fundamental link between corruption and bank credit risk.
- Identifies 'Bank Type' (Conventional vs. Islamic) as a critical variable to include in our model for the South Asian region.

Phan Huy, T., Nguyen Thi Hai, Y., Doan Thi Ngoc, D., & Bui Nguyen Thuy, N. (2025). Corruption, credit risk and economic growth: Global empirical evidence. *Southeast Asian Journal of Economics*, 13(2), 1-36.

- Key Findings:

- A global study of 218 countries confirms that corruption increases a bank's NPLs and weakens the institution.
- The negative impact of corruption on credit risk is more severe in developing countries that have weaker regulatory frameworks.

- Relevance to Our Study:

- Offers large-scale evidence confirming our core hypothesis.
- Justifies the specific regional focus on South Asia, as its developing economies are more vulnerable to the effects of corruption

Hasan, R., & Ashfaq, M. (2021). Corruption and its diverse effect on credit risk: global evidence. *Future Business Journal*, 7(1), 18.

Key Findings

- **Global Positive Correlation:** There is a direct and statistically significant link between higher national corruption levels and increased credit risk.
- **Impact on Banking:** Generally, countries with more corruption tend to have a higher percentage of non-performing loans, indicating greater instability in their banking systems.

Relevance to our Research

- **Provides a Testable Hypothesis:** The "Grease the Wheels" hypothesis offers a specific framework to test, suggesting the relationship between corruption and credit risk is not always linear.
- **Context-Dependent Relationship:** It proposes that the effect of corruption changes based on the existing level of corruption. While corruption increases credit risk in low-corruption countries, it may actually *decrease* credit risk in highly corrupt countries by serving as a way to bypass inefficient bureaucracy.

Corruption and Credit Risk

Hypothesis (H1):

H₀: Corruption has no significant effect on the NPL ratio.

H₁: Higher corruption levels are associated with higher NPL ratios.

Rationale:

Corruption reduces institutional quality and weakens enforcement of banking regulations. In corrupt environments, bank officers may grant loans based on favoritism or bribes instead of creditworthiness, increasing the likelihood of default. Prior studies (e.g., Park, 2021; Louzis et al., 2012) find that countries with weak governance and higher corruption levels tend to have higher NPLs.

Hence, a positive relationship between corruption and NPLs is expected.



Economic Growth and Credit Risk

Hypothesis (H2):

H_0 : GDP growth has no significant effect on NPL ratios.

H_1 : Higher GDP growth rates lead to lower NPL ratios.

Rationale:

In periods of strong economic growth, borrower income and firm profitability rise, improving repayment capacity and reducing loan defaults. Conversely, during recessions, loan performance deteriorates as firms struggle to meet obligations. Therefore, economic expansion is expected to lower the level of NPLs (negative relationship).



Inflation and Credit Risk

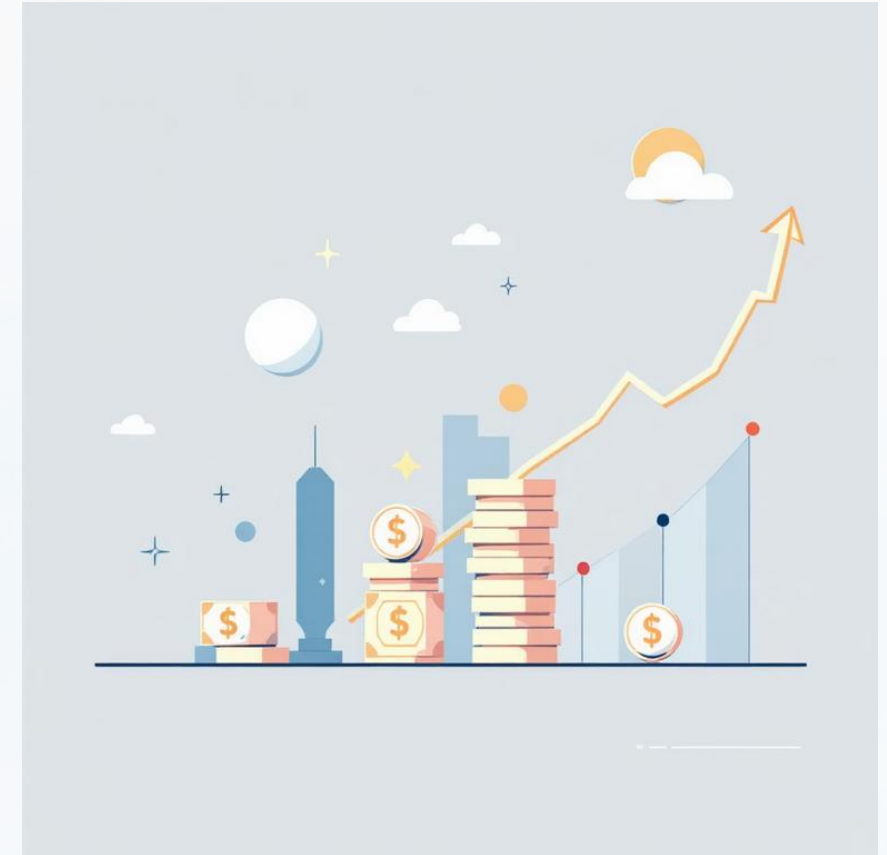
Hypothesis (H3):

H_0 : Inflation has no significant effect on NPL ratios.

H_1 : Higher inflation rates increase NPL ratios.

Rationale:

Inflation reduces the real value of income and assets, worsening borrowers' ability to service debt. Moreover, high inflation may distort interest rate structures, leading to higher loan defaults. In emerging economies like those in South Asia, inflation also reflects macroeconomic instability, which further deteriorates credit quality. Thus, a positive relationship is anticipated.



Bank Size and Credit Risk

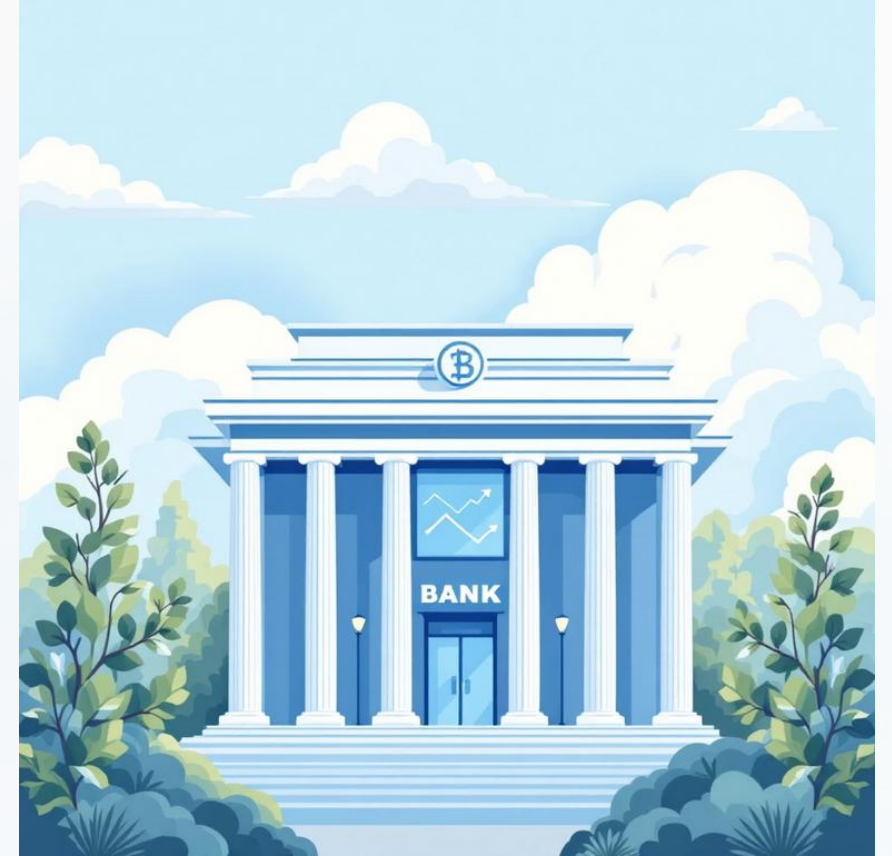
Hypothesis (H4a):

H₀: Bank size has no significant effect on NPL ratios.

H₁: Larger banks have lower NPL ratios.

Rationale:

Larger banks generally benefit from diversification, better risk management, and stronger internal controls. They can absorb credit shocks more efficiently than smaller banks. Thus, as bank size (measured by log of total assets) increases, the NPL ratio is expected to decline (negative sign).



Profitability (ROE) and Credit Risk

Hypothesis (H4b):

H₀: Profitability has no significant effect on NPL ratios.

H₁: Higher profitability (ROE) leads to lower NPL ratios.

Rationale:

A profitable bank indicates efficient management and sound credit practices. Banks with higher ROE typically manage lending portfolios more effectively, ensuring credit quality and timely recoveries. Profitability can therefore serve as an indicator of prudent lending, implying a negative relationship with NPLs.



Credit Risk Persistence (Lagged NPLs)

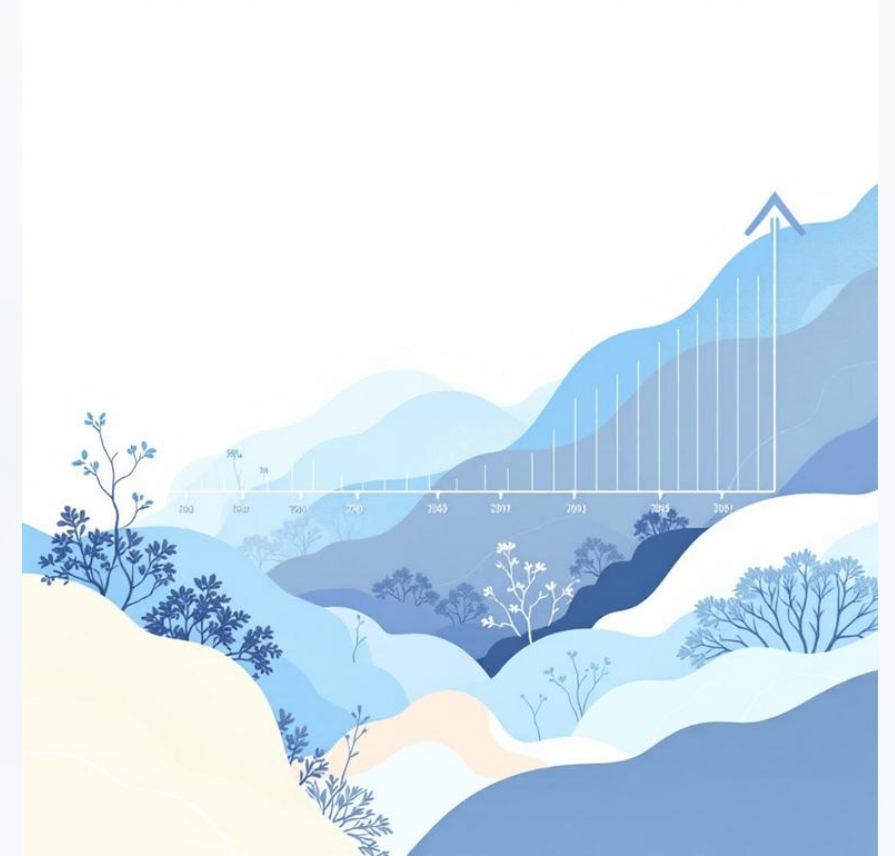
Hypothesis (H5):

H_0 : The previous year's NPL ratio has no effect on the current NPL ratio.

H_1 : The previous year's NPL ratio positively affects the current NPL ratio.

Rationale:

NPLs tend to be persistent over time because bad loans are carried forward and take multiple years to resolve. A bank with high NPLs in the previous year likely continues facing similar issues in subsequent years, showing temporal dependence. Thus, the lagged NPL variable is expected to have a positive and significant coefficient.



Stability of the CPI-NPL Relationship Across Countries

Null Hypothesis (H_0):

The effect of the Corruption Index on NPL Ratios is consistent across all countries.

Alternative Hypothesis (H_1):

The effect of the Corruption Index on NPL Ratios is not consistent across all countries.



Research Design

This study employs a structured, quantitative approach to investigate the factors influencing Non-Performing Loans (NPLs) in South Asian banking systems from 2014 to 2024. The design is twofold: first, an **explanatory econometric analysis** to test hypotheses regarding the determinants of NPLs, and second, a **predictive machine learning analysis** to build and evaluate models for forecasting NPL ratios.

Research Philosophy: Quantitative, Positivist

Strategy: Explanatory & Predictive

Approach: Longitudinal Panel Study (2014-2024)

Core Methods: Econometric Panel Regression & Predictive Machine Learning

Analytical Framework (Part 1): Econometric Analysis

Goal: To **explain** the drivers of NPLs and test hypotheses.

Step 1: Diagnostics

Conducted Exploratory Data Analysis (EDA) to understand data patterns.

Ran Variance Inflation Factor (VIF) test; found **no multicollinearity**.

Step 2: Model Selection

Compared Pooled OLS, Fixed Effects (FE), and Random Effects (RE) models.

The **Hausman Test** ($p=0.18$) confirmed the **Random Effects (RE)** model as the most appropriate choice.

Step 3: Final Model

A **Random Effects model with Time Dummies** was used for final hypothesis testing to account for both bank- and year-specific effects.

Analytical Framework (Part 2): Predictive Modeling

Goal: To predict NPL ratios with high accuracy.

Models Tested:

- Random Forest
- Gradient Boosting
- XGBoost

Process:

- Used **GridSearchCV** with **TimeSeriesSplit** for robust hyperparameter tuning .
- Evaluated models on a held-out test set using R^2 , RMSE, and MAE.

Outcome:

- Gradient Boosting** was the top-performing model, explaining **76.1%** of the variance in NPLs on unseen data.

Data Collection Methodology

Data Source: Secondary Data

- Bank Annual Reports
- World Bank Open Data
- Transparency International

Sampling: Purposive Convenience
Sampling

Scope & Scale:

- 7 South Asian Countries
- 67 Commercial Banks
- **11-Year** Period (2014-2024)



Variable Category	Variable Name	Measurement	Source	Supporting Literature
Target Variable	NPL Ratio	Non-Performing Loans / Total Loans (%).	Bank Annual Report	Serengil et al. (2021)
Key Predictor	Corruption Index	100 - CPI Score.	Transparency Org	Rehman et al. (2020)
Macroeconomic Variables	GDP Growth	Annual GDP Growth (%).	World Bank Open Data	Louzis et al. (2012)
	Inflation (INF)	Consumer Price Index, annual % change.	World Bank Open Data	Louzis et al. (2012)
	Population Growth (POP)	Annual population growth (%).	World Bank Open Data	Louzis et al. (2012)
Bank-Specific Variables	Bank Size (SIZE)	Natural Logarithm of Total Assets.	Bank Annual Report	Alhassan et al. (2014)
	Return on Assets (ROAA)	Net Income / Total Assets.	Bank Annual Report	Masrom et al. (2023);
	Return on Equity (ROAE)	Net Income / Shareholders' Equity.	Bank Annual Report	Masrom et al. (2023);
	Age	Current Year - Year of Establishment.	Bank Incorporation	Derived concept
	NPL_lag	The previous year's NPL ratio.	Bank Annual Report	Than et al. (2025)

Dataset

Structure: A panel dataset of **733** bank-year observations.

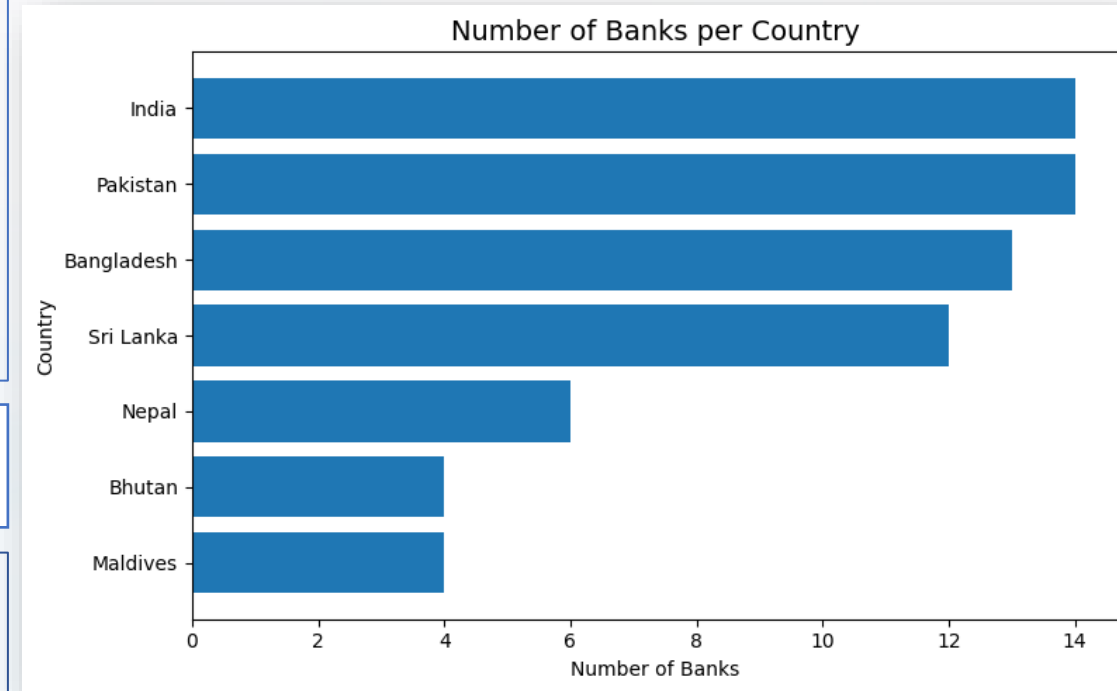
Scope: **67** unique commercial banks across **7** South Asian countries (India, Pakistan, Bangladesh, Sri Lanka, Nepal, Bhutan, Maldives)

Period: 2014 to 2024.

Key Variable

Dependent Variable: NPL Ratio (log-transformed to normalize distribution).

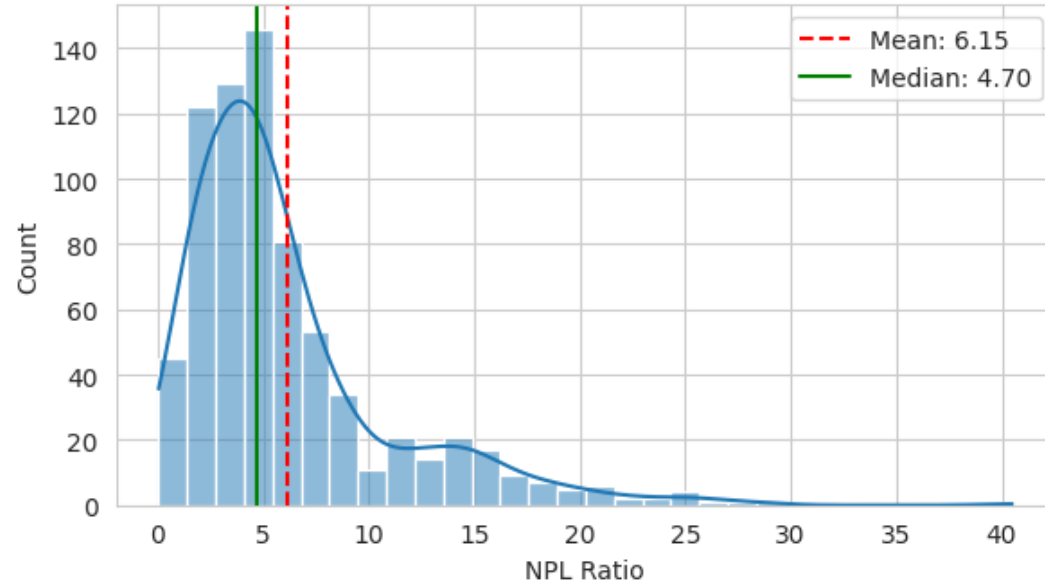
Independent Variables: GDP Growth, Inflation, Corruption Index, Bank Size (log of assets), Return on Equity (ROE), and the NPL Ratio from the previous year (Lagged NPL).



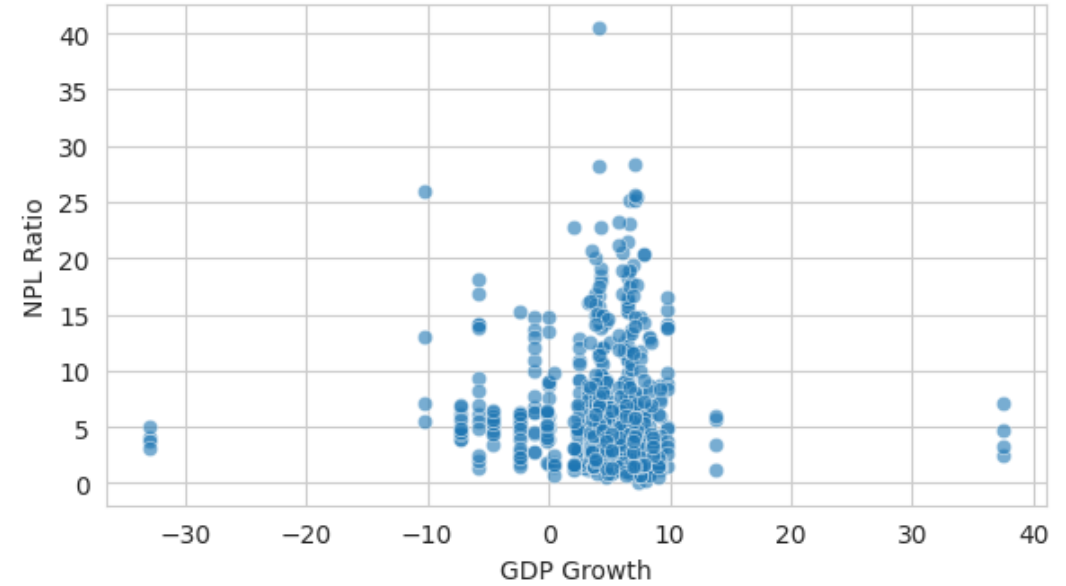
Variable	Mean	Standard Dev.	Min.	25%	50%	75%	Max.
NPL Ratio	6.147	5.044	0.05	2.87	4.7	7.37	40.49
Corruption Index	66.954	5.379	57	62	67	72	77
GDP Growth	4.594	5.285	-32.9	3.7	5.8	6.9	37.5
Inflation (INF)	7.128	7.205	-1.6	3.9	5.6	7.7	49.72
Population Growth	1.082	0.681	-0.651	0.8	1.006	1.403	3.5
Bank Size (SIZE)	20.266	5.254	6.71	20.346	22.081	23.052	27.42
ROA	1.225	1.318	-3.88	0.65	1.11	1.65	22.2
ROE	12.647	9.812	-51.53	8.66	13.8	17.35	56.27
Age	47.317	38.233	0	21	32	65	170
NPL_lag	6.258	5.109	0.05	2.9	4.72	7.53	32.77

Exploratory Data Analysis of NPL Ratio Determinants

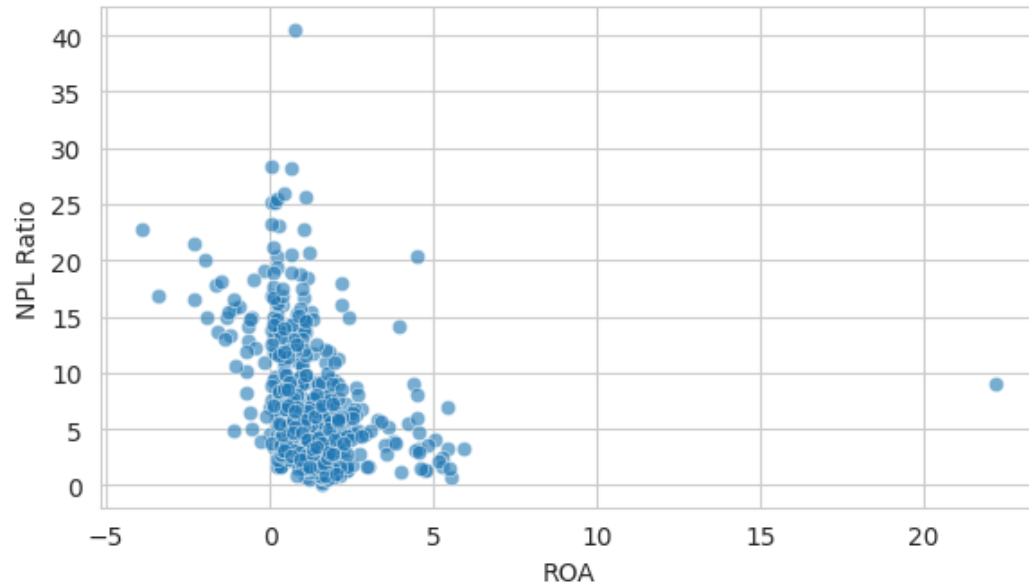
1. Distribution of NPL Ratio



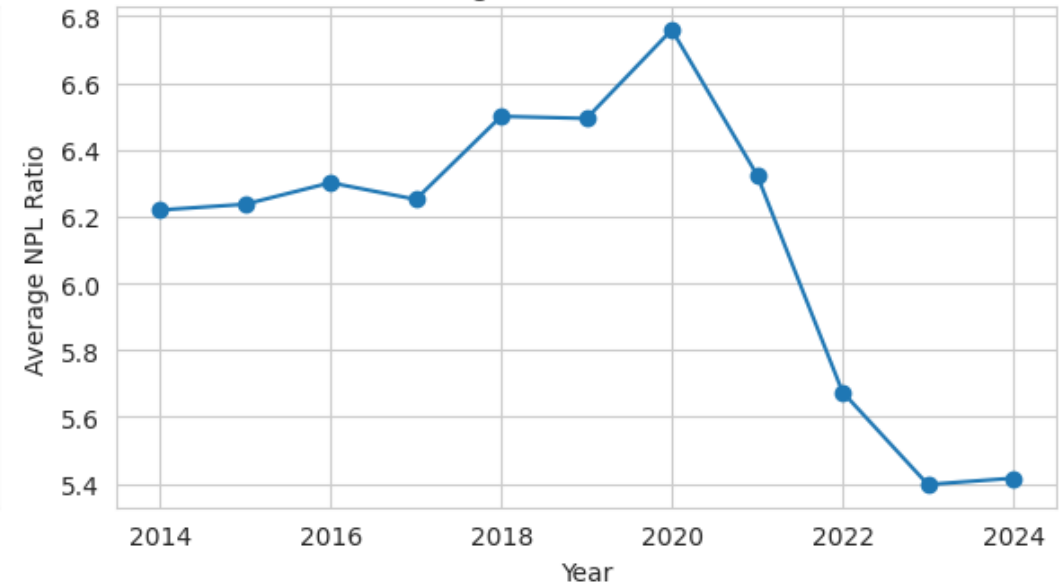
2. NPL Ratio vs. GDP Growth



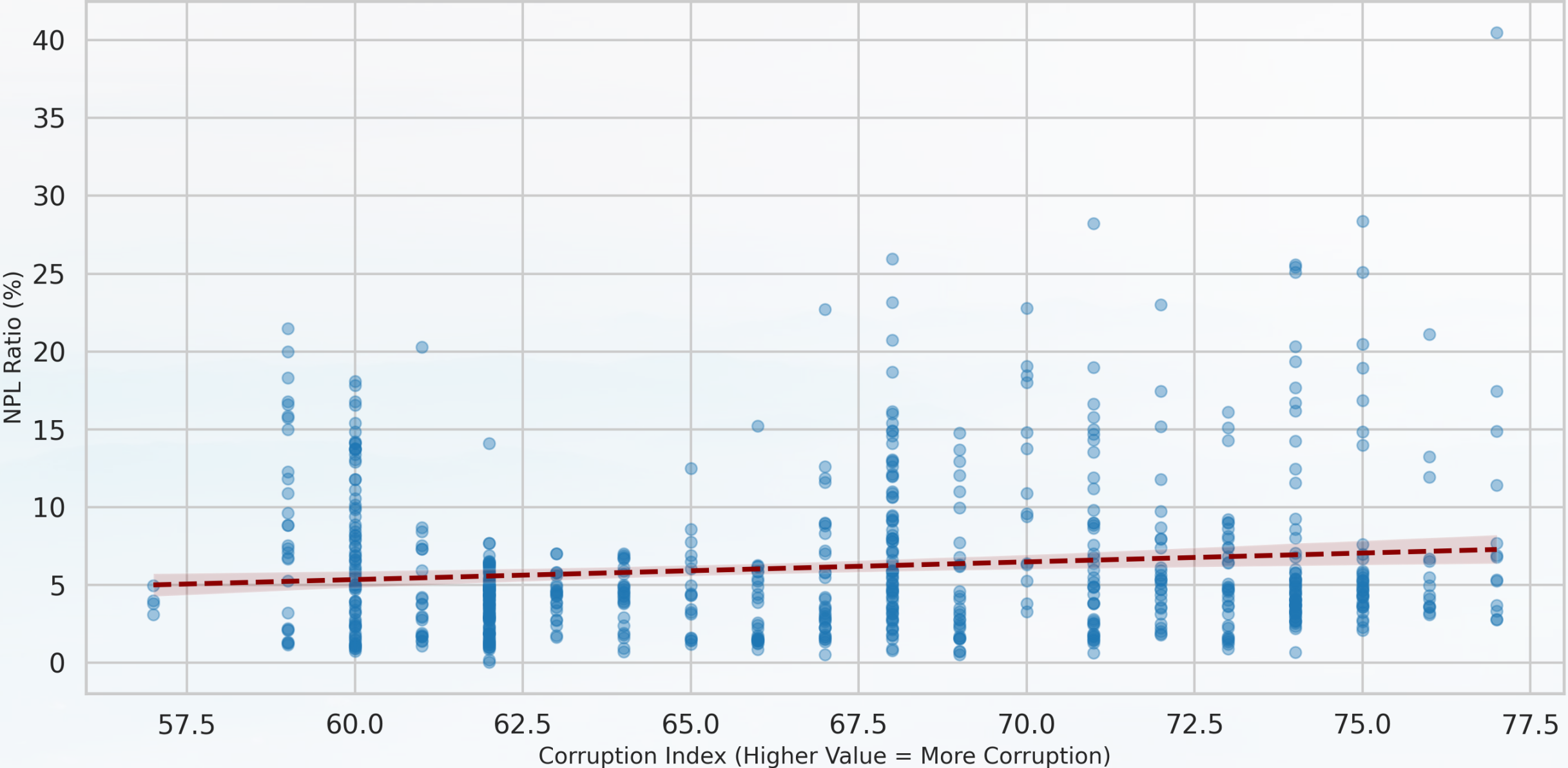
3. NPL Ratio vs. ROA



4. Average NPL Ratio Over Time

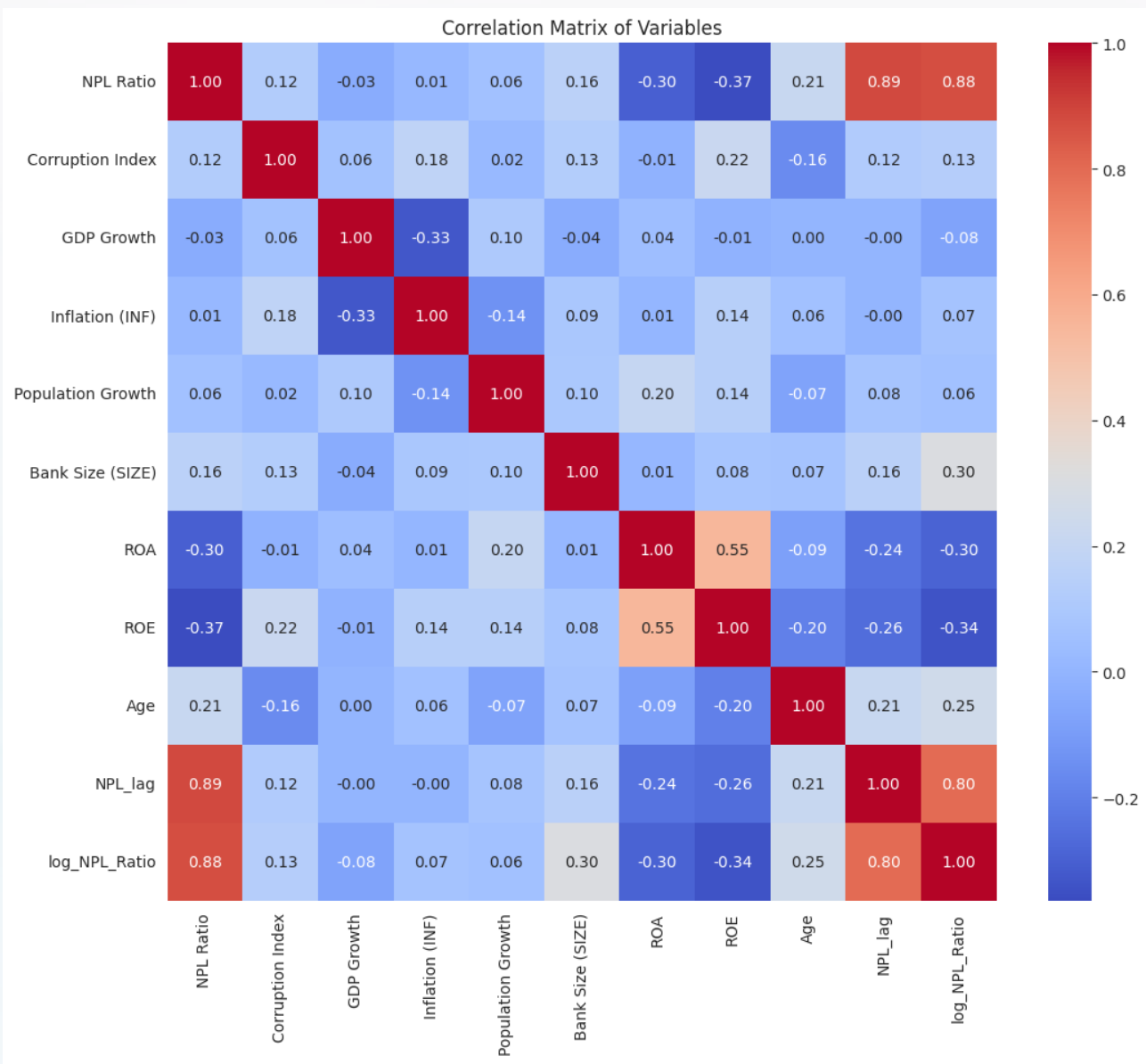


Overall Relationship Between Corruption Index and NPL Ratio (All Countries Pooled)



EDA Interpretation

- 1. Distribution of NPL Ratio:** The histogram clearly shows the data is **heavily right-skewed**. The mean (6.15) is pulled to the right of the median (4.70) by the long tail of high NPL values. For linear regression models to work best, the target variable should be more symmetrically distributed.
- 2. NPL Ratio vs. GDP Growth:** The relationship isn't a simple straight line. The highest NPLs seem to occur during periods of moderate, rather than extremely low or high, GDP growth. This suggests a potentially complex, non-linear relationship.
- 3. NPL Ratio vs. ROA:** This shows a clear and expected **negative relationship**. As bank profitability (ROA) increases, the NPL Ratio tends to decrease significantly. This suggests ROA will be a strong predictor.
- 4. Average NPL Ratio Over Time:** We can see a distinct time trend. The average NPL Ratio peaked around 2020—likely due to the economic shock of the pandemic—and then fell sharply in the following years. This confirms that year-specific effects are important.



Variable	Correlation Coefficient
log_NPL_Ratio	1
NPL Ratio	0.8764
NPL_lag	0.7998
ROE	-0.343
Bank Size (SIZE)	0.3047
ROA	-0.2956
Age	0.2529
Corruption Index	0.1332
GDP Growth	-0.077
Inflation (INF)	0.0709
Population Growth	0.0607

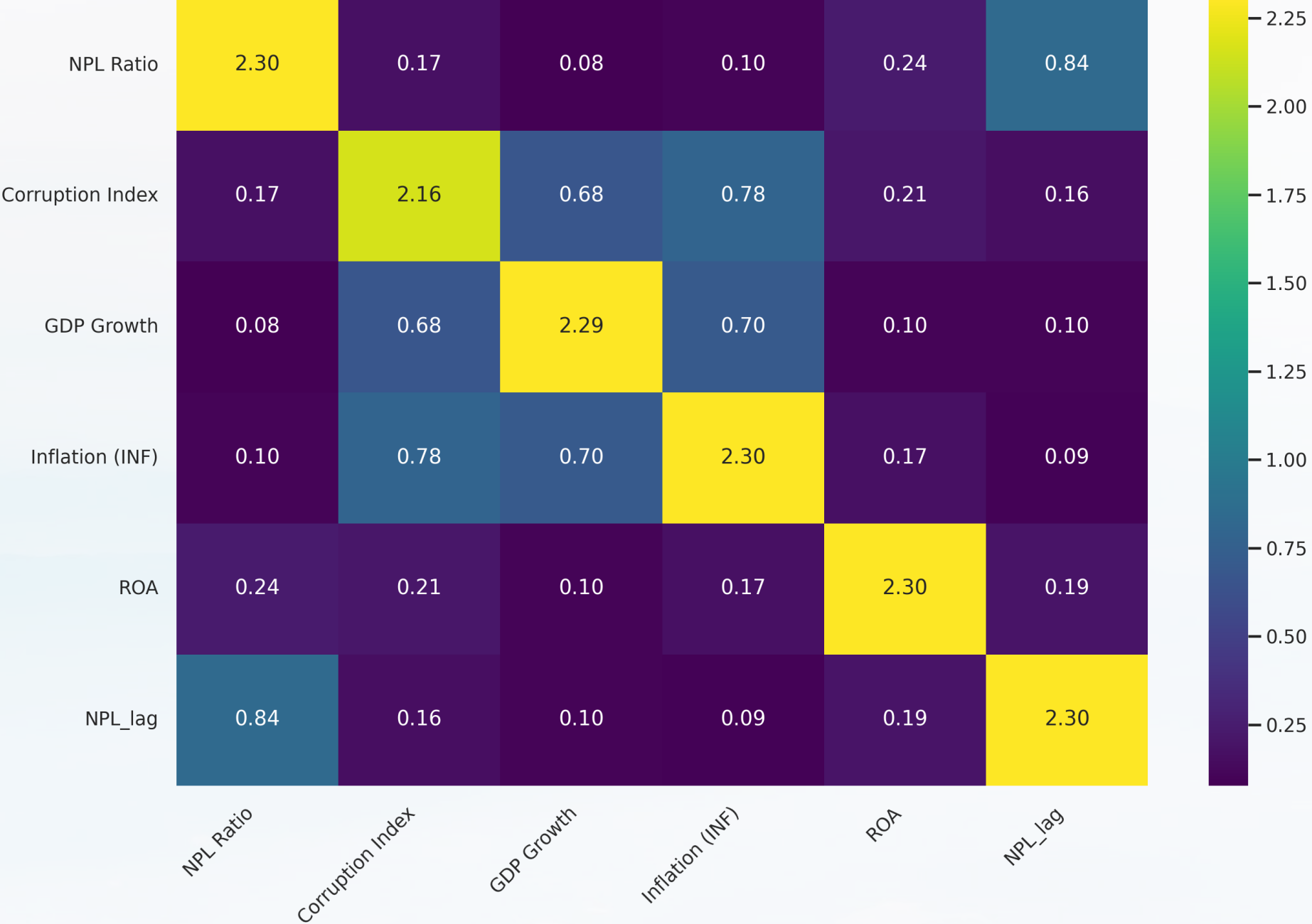
Strongest Predictors:
NPL_lag (0.80)

ROE (-0.34) & ROA (-0.30)

Bank Size (SIZE) (0.30)

Potential Multicollinearity:
0.55 correlation between ROA and ROE.

Overall Mutual Information Matrix for All Variables



Multicollinearity Check (VIF)

We've spotted a potential issue visually, and the next step is to test for it formally using the **Variance Inflation Factor (VIF)**.

Feature	VIF Score
ROE	1.6772
ROA	1.5346
Inflation (INF)	1.2301
NPL_lag	1.2177
Corruption Index	1.1949
GDP Growth	1.1551
Age	1.1303
Population Growth	1.1074
Bank Size (SIZE)	1.0676

Analysis of the VIF Scores

No Multicollinearity: The highest VIF score is approximately 1.68 for ROE. All scores are **well below the common threshold of 5**.

Keep All Variables: This result tells us that multicollinearity is not an issue in your dataset.

Panel Data Regression Models

We ran three key models to analyze the determinants of the log_NPL_Ratio. This approach allowed us to see how controlling for the unique characteristics of each bank affects the results.

1. **Pooled OLS:** A basic regression that ignores the panel structure (i.e., it treats all 733 observations as independent). It serves as a simple baseline.
2. **Fixed Effects (FE):** This model controls for all unobserved, time-invariant characteristics of each bank (e.g., its business model, risk culture, management quality).
3. **Random Effects (RE):** This model assumes the unobserved bank-specific characteristics are random and uncorrelated with the other predictors.
4. **Hausman Test:** After running the FE and RE models, we will use this test to formally decide which one is more appropriate for our data.

Pooled OLS Regression: Determinants of NPL Ratio

This analysis examines the factors influencing the Non-Performing Loan (NPL) Ratio using a Pooled OLS model.

Statistic	Value
Dependent Variable	log_NPL_Ratio
R-squared	0.717
Observations (N)	733
F-statistic	50.8
Prob (F-statistic)	0

Variable	Coefficient	Std. Error	P-value
NPL_lag	0.1072	0.007	0.000***
Bank Size (SIZE)	0.0282	0.007	0.000***
ROE	-0.0141	0.003	0.000***
Inflation (INF)	0.0052	0.002	0.005***
GDP Growth	-0.0093	0.004	0.024**
Corruption Index	0.0102	0.007	0.119
Population Growth	0.0451	0.031	0.145
Age	0.0013	0.001	0.15
ROA	-0.0213	0.023	0.358
Constant	-0.3009	0.448	0.502

The pooled OLS model assumes identical intercepts across all banks and years. The results show statistically significant coefficients for GDP growth, Inflation, and ROE.

However, the model's **F-test for poolability ($p < 0.001$)** rejects the null of homogeneous intercepts, indicating the need for a panel model that accounts for heterogeneity.

While the pooled OLS identifies basic linear relationships, it fails to capture cross-bank and cross-year differences, thus serving only as a diagnostic benchmark.

Fixed Effects (FE) Model Results

This model controls for stable, unobserved characteristics unique to each bank (entity effects) to analyze how variables changing *over time* affect the NPL Ratio.

Statistic	Value
Dependent Variable	log_NPL_Ratio
R-squared (Within)	0.449
Observations (N)	733
Number of Entities	67
F-statistic	23.78
Prob (F-statistic)	0.0000

Variable	Coefficient	Std. Error	P-value
NPL_lag	0.0842	0.008	0.000***
ROE	-0.0184	0.005	0.000***
GDP Growth	-0.0066	0.003	0.025**
Inflation (INF)	0.0027	0.001	0.047**
Corruption Index	0.0026	0.011	0.806
Bank Size (SIZE)	0.0110	0.016	0.486
Population Growth	-0.0141	0.050	0.779
ROA	-0.0039	0.016	0.810
Age	-0.0010	0.007	0.879

The fixed effects model introduces bank-specific intercepts, controlling for all time-invariant characteristics.

Key findings:

- **GDP Growth (-0.0066 , $p < 0.05$):** Economic growth significantly reduces NPLs, consistent with macroeconomic stability improving loan performance.
- **Inflation ($+0.0027$, $p < 0.05$):** Inflation increases NPLs, reflecting erosion of real repayment capacity.
- **ROE (-0.0184 , $p < 0.001$):** More profitable banks face lower credit risk.
- **NPL_lag ($+0.084$, $p < 0.001$):** Strong persistence — previous-year NPLs are a strong predictor of current NPLs.

The **within $R^2 = 0.44$** indicates that nearly half of the within-bank variation in NPL ratios is explained by these factors.

Interpretation:

The FE results support the hypothesis that macroeconomic conditions and profitability significantly influence bank credit risk, with persistent effects over time.

Random Effects (RE) Model Results

The Random Effects model acts as a compromise between Pooled OLS and Fixed Effects, assuming that the unique characteristics of each bank are random and not correlated with the predictor variables.

Statistic	Value
Dependent Variable	log_NPL_Ratio
R-squared (Overall)	0.709
Observations (N)	733
Number of Entities	67
F-statistic	40.96
Prob (F-statistic)	0.0000

Variable	Coefficient	Std. Error	P-value
NPL_lag	0.0946	0.006	0.000***
ROE	-0.0171	0.004	0.000***
Bank Size (SIZE)	0.0297	0.008	0.000***
GDP Growth	-0.0072	0.003	0.018**
Inflation (INF)	0.0033	0.001	0.027**
Corruption Index	0.0102	0.006	0.091*
Age	0.0015	0.001	0.134
Population Growth	0.0239	0.037	0.518
ROA	-0.0108	0.019	0.573

Key Takeaways & Comparison

Similar to FE: The core predictors **NPL_lag**, **ROE**, **GDP Growth**, and **Inflation** remain significant, showing their robust impact on NPLs.

Different from FE:

Bank Size is once again highly significant, similar to the Pooled OLS results. The RE model suggests that stable differences between banks' sizes are important predictors.

Corruption Index is now significant at the 10% level, suggesting a weak positive relationship with NPLs.

Model Selection: Hausman Test

To determine whether the Fixed Effects or Random Effects model is more appropriate for the analysis, we use the Hausman test.

The Core Question

The test helps us decide if the unobserved unique characteristics of each bank (the "effects") are correlated with the other variables in the model.

- **Null Hypothesis (H_0):** The effects are random and uncorrelated. The **Random Effects model is preferred**.
- **Alternative Hypothesis (H_1):** The effects are correlated. The **Fixed Effects model is preferred**.

Statistic	Value
Chi-Squared (X2)	13.84
P-value	0.1803

- **Decision Rule:** We compare the p-value to our significance level (typically).
- **Interpretation:** Since the **p-value (0.18)** is greater than **0.05**, we fail to reject the null hypothesis.
- **Final Verdict:** There is not enough statistical evidence to suggest that the Fixed Effects model is necessary. Therefore, the **Random Effects (RE) model is the most appropriate and efficient choice** for your final analysis.

Random Effects model with Time-effects

This model is an improvement on the previous one as it includes **year-specific dummy variables** to control for factors that affected all banks in a particular year (e.g., broad economic shocks or policy changes).

Statistic	Value
Dependent Variable	log_NPL_Ratio
R-squared (Overall)	0.710
Observations (N)	733
F-statistic	24.81
Prob (F-statistic)	0.0000

Variable	Coefficient	P-value
NPL_lag	0.0936	0.000***
ROE	-0.0171	0.000***
Bank Size (SIZE)	0.0297	0.000***
Inflation (INF)	0.0063	0.001***
Age	0.0015	0.137
Corruption Index	0.0103	0.107
GDP Growth	-0.0051	0.166
ROA	-0.0070	0.721
Population Growth	0.0053	0.896

Key Takeaways & Final Interpretation

Robust Predictors: The core drivers of NPLs remain highly significant and consistent. **Past NPLs**, **ROE** (profitability), **Bank Size**, and **Inflation** are the most robust predictors in your final model.

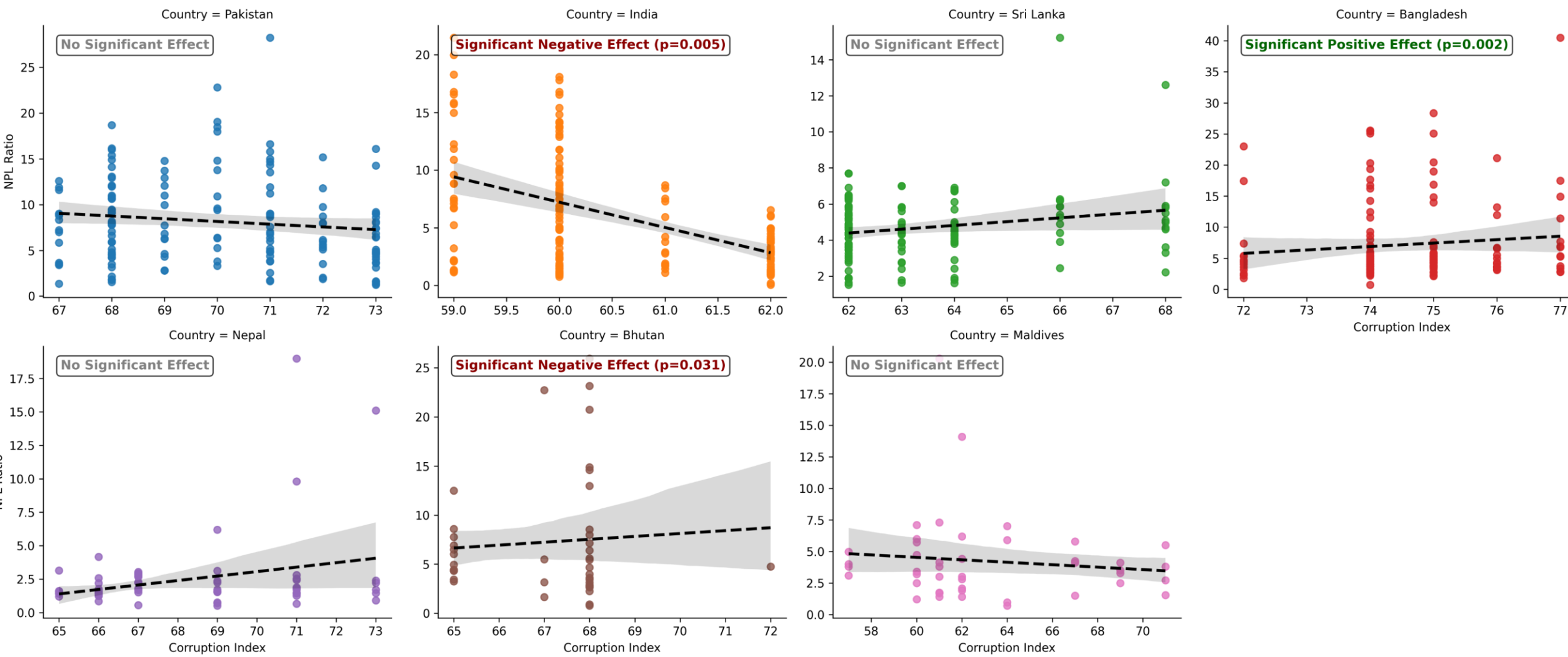
The Impact of Time Dummies:

GDP Growth is no longer significant. This is a critical finding. It suggests that the effect of GDP growth seen in earlier models was actually capturing broader year-to-year economic trends. Now that those trends are controlled for by the year dummies, the specific effect of GDP growth disappears.

The year dummies themselves are not significant (all have p-values > 0.10). This indicates that, after accounting for all the other variables, no single year had a statistically significant, unique impact on NPLs compared to the base year (2014).

Final Conclusion: The analysis strongly suggests that a bank's NPL ratio is primarily driven by its **own past performance (inertia)**, its **profitability**, and its **size**, along with its sensitivity to broad inflation. The impact of the general business cycle (GDP) is less direct than previously thought.

Country-Specific Relationship Between Corruption Index and NPL Ratios



Final Verdict on the Hypothesis of Consistency

Hypothesis Tested: A Chow Test was used to evaluate the null hypothesis (H_0) that the relationship between the Corruption Index and NPL Ratios is consistent across all seven countries.

Statistical Result: The test yielded a highly significant result ($p\text{-value} < 0.0001$), falling well below the standard 0.05 significance threshold.

Conclusion: The null hypothesis is formally rejected. This provides conclusive statistical evidence that the relationships are inconsistent, validating our country-by-country analytical approach.

Evaluation of Hypotheses:

The following section evaluates the initial research hypotheses based on the final econometric model: the **Random Effects model with Time Dummies**. Each hypothesis is assessed against the model's coefficient estimates and their statistical significance.

H1: Corruption and Credit Risk

Hypothesis (H_1): Higher corruption levels are associated with higher NPL ratios.

Result: The model produced a positive coefficient for the Corruption Index, which aligns with the hypothesis. However, the effect was not statistically significant at conventional levels (p-value = 0.107).

Conclusion: The null hypothesis (H_0) is not rejected. The analysis does not provide sufficient statistical evidence to support the claim that corruption has a significant effect on the NPL ratio in this context.

H2: Economic Growth and Credit Risk

Hypothesis (H_1): Higher GDP growth rates lead to lower NPL ratios.

Result: The coefficient for GDP Growth was negative as hypothesized, but it failed to achieve statistical significance (p-value = 0.166). This lack of significance emerged after controlling for year-fixed effects.

Conclusion: The null hypothesis (H_0) is not rejected. The effect of general economic trends appears to be captured by the time dummies, rendering the specific impact of GDP growth insignificant.

H3: Inflation and Credit Risk

Hypothesis (H₁): Higher inflation rates increase NPL ratios.

Result: The analysis revealed a positive and highly statistically significant relationship between the inflation rate and the NPL ratio (p-value = 0.001).

Conclusion: The null hypothesis (H₀) is rejected. The data provides strong support for the alternative hypothesis (H₁), confirming that higher inflation is associated with increased credit risk.

H4a: Bank Size and Credit Risk

Hypothesis (H₁): Larger banks have lower NPL ratios.

Result: The model found a highly significant relationship between bank size and NPLs (p-value < 0.001). However, the coefficient was positive, indicating that larger banks are associated with *higher* NPL ratios.

Conclusion: The null hypothesis (H₀) is rejected, but the evidence **contradicts** the alternative hypothesis (H₁). This is a significant finding that runs counter to the initial expectation.

H4b: Profitability and Credit Risk

Hypothesis (H₁): Higher profitability leads to lower NPL ratios.

Result: The specific measure of profitability, Return on Assets (ROA), was not statistically significant (p-value = 0.721). However, the alternative measure, **Return on Equity (ROE)**, demonstrated a negative and highly significant relationship with NPLs (p-value < 0.001).

Conclusion: The underlying hypothesis is **supported** through the ROE metric. Higher profitability, as measured by ROE, is a significant factor in reducing credit risk.

H5: Credit Risk Persistence (Lagged NPLs)

Hypothesis (H₁): The previous year's NPL ratio positively affects the current NPL ratio.

Result: The coefficient for the lagged NPL ratio was positive and highly statistically significant (p-value < 0.001).

Conclusion: The null hypothesis (H₀) is rejected. The data provides robust evidence in support of H₁, confirming the persistent nature of credit risk.

H6: Stability of the CPI-NPL Relationship Across Countries

Hypothesis (H₁): The effect of the Corruption Index on NPL Ratios is not consistent across all countries.

Result: The test yielded a highly significant result (p-value < 0.0001), falling well below the standard 0.05 significance threshold.

Conclusion: The null hypothesis is formally rejected. This provides conclusive statistical evidence that the relationships are inconsistent, validating our country-by-country analytical approach.

Summary Table of Results

Hypothesis	Expected Sign	Estimated Sign	P-value	Conclusion
H1: Corruption	Positive (+)	Positive (+)	0.107	Not Supported
H2: GDP Growth	Negative (-)	Negative (-)	0.166	Not Supported
H3: Inflation	Positive (+)	Positive (+)	0.001	Supported
H4a: Bank Size	Negative (-)	Positive (+)	< 0.001	Contradicted
H4b: Profitability	Negative (-)	Supported (ROE)	< 0.001	Supported (via ROE)
H5: NPL Persistence	Positive (+)	Positive (+)	< 0.001	Supported

Predictive Model Performance Summary

This report summarizes the performance of three machine learning models (Random Forest, Gradient Boosting, and XGBoost) developed for a regression task. The models were optimized using GridSearchCV and evaluated on a final, held-out test set to assess their predictive accuracy on unseen data.

Model	Test R ²	CV R ² (Train)	Test RMSE	Test MAE
Gradient Boosting	0.761	0.783	0.344	0.257
XGBoost	0.735	0.745	0.363	0.271
Random Forest	0.716	0.775	0.375	0.284

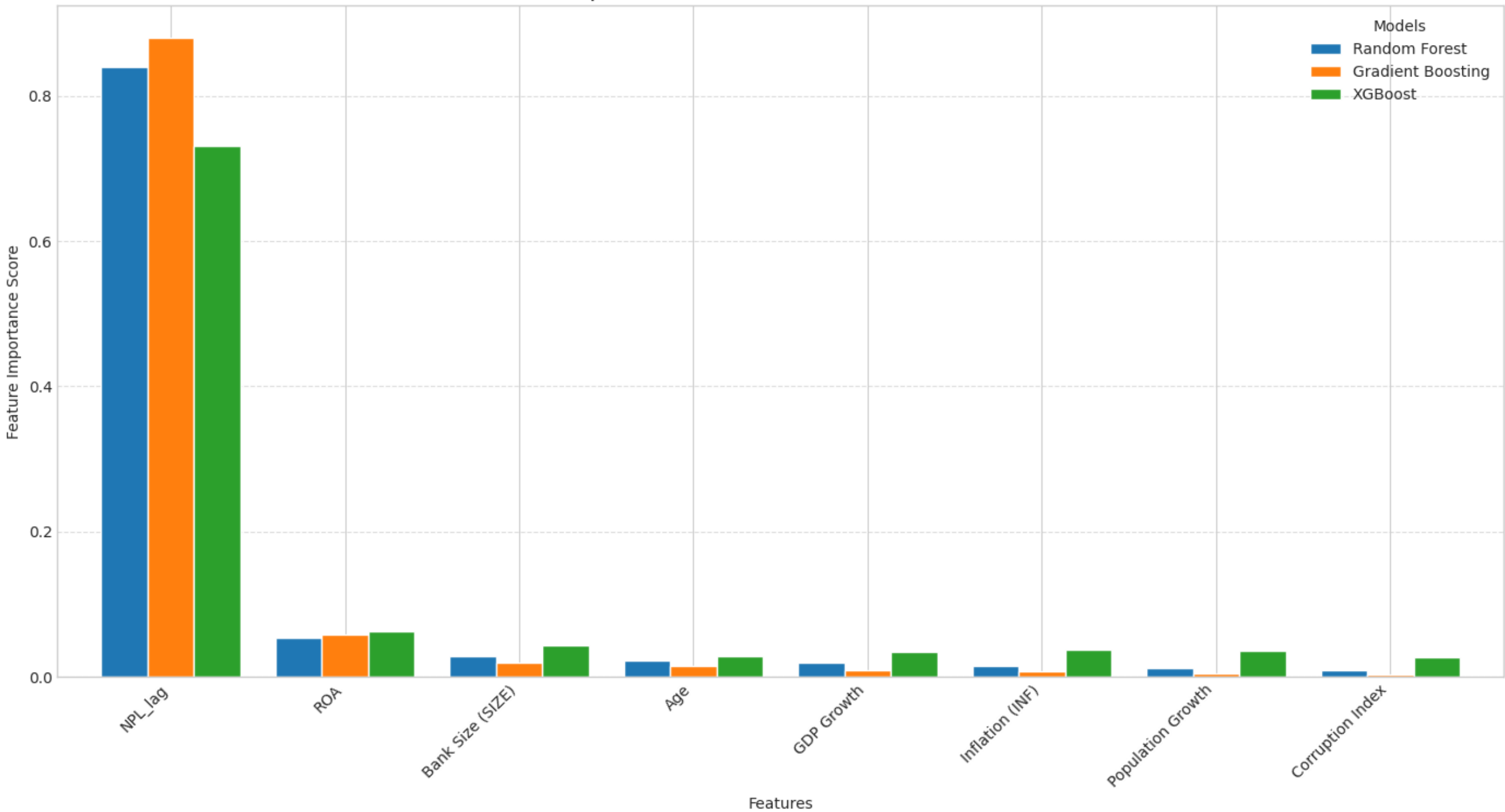
Top Performing Model: The **Gradient Boosting** model is the clear winner, achieving the best performance across all evaluation metrics, including the highest R-squared (0.761) and the lowest error rates (RMSE and MAE).

Strong Predictive Power: The final Gradient Boosting model is highly effective, explaining **76.1%** of the variance in the target variable on the test set. This indicates a strong level of predictive accuracy suitable for practical application.

Excellent Generalization: There is a minimal gap between the model's cross-validation score on the training data (CV R² = 0.783) and its final performance on the test data (Test R² = 0.761). This suggests the model is robust, generalizes well to new data, and is not significantly overfit.

Conclusion: The optimized **Gradient Boosting model is the recommended choice** for this predictive task due to its superior accuracy and reliable generalization performance.

Feature Importance Across Models for NPL Prediction



Limitations

Omitted Variable Bias: The models do not include all potential factors influencing NPLs, such as interest rates, regulatory quality, or bank-specific capital ratios. This could affect the accuracy of the results.

Potential for Dynamic Panel Bias (Nickell Bias): The inclusion of a lagged dependent variable (NPL_lag) in a panel model with a relatively short time series (11 years) can introduce "Nickell bias".

Measurement of Corruption: The research relies on the **Corruption Perceptions Index (CPI)**, which measures the *perception* of corruption rather than actual instances of corrupt transactions.

Sampling Method and Generalizability: The study uses **purposive convenience sampling**. Because this is a non-probability sampling method, the findings may not be fully generalizable to all commercial banks in South Asia, particularly smaller or unlisted banks whose data is less accessible.

Data Consistency: The definition and accounting standards for NPLs can vary across jurisdictions.

Non-Performing Loans (NPLs): The official NPL ratios themselves can be unreliable. Banks, especially in environments with weak regulation, may engage in "evergreening" (restructuring loans to troubled borrowers to keep them classified as "performing"), thus artificially lowering the reported NPL numbers.

Linearity Assumption: The core econometric models (OLS, FE, RE) all assume a linear relationship between the predictors and the log of NPLs. The real-world effect could be non-linear.

Conclusion of the Study

The central hypothesis is decisively rejected. The relationship between corruption and Non-Performing Loan (NPL) ratios is not consistent across the seven South Asian nations studied; it is highly dependent on the specific country.

The effect of corruption on NPLs manifested in three distinct, contradictory ways:

- A "**Sand in the Wheels**" effect was found in Bangladesh, where higher corruption is significantly linked to higher NPLs.
- A "**Grease the Wheels**" effect was found in India and Bhutan, where higher corruption is significantly linked to lower officially reported NPLs.
- No significant effect was found in Pakistan, Sri Lanka, Nepal, and the Maldives.

Conclusion of the Study

1. **Credit Risk is Highly Persistent:** The most powerful predictor of a bank's NPL ratio is its NPL ratio from the previous year. This confirms that credit risk has significant inertia and bad loans take time to resolve.
2. **Key Drivers Identified:** Beyond persistence, a bank's credit risk is significantly driven by its **profitability (ROE)**, **size**, and the macroeconomic pressure of **inflation**. More profitable banks manage risk better, while higher inflation and larger size were associated with higher NPLs.
3. **High Predictive Accuracy Achieved:** Machine learning models proved highly effective. The **Gradient Boosting model** was the top performer, successfully explaining **76.1%** of the variance in NPLs on unseen data, demonstrating strong potential for practical risk forecasting.

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